



2.1 Population Distribution & Density

Objective and Essential Learning

2.1.1 Identify the factors that influence the distribution of population at different scales.

- A1. Physical features (e.g. climate, landforms, water bodies) and human factors (e.g. culture, economics, history, politics) influence the distribution of population.
- A2. Factors that illustrate patterns of population distribution vary according to the scale of analysis.



2.1.2 Define and explain the differences between the methods geographers use to calculate population density.

- B. The three methods for calculating population density are arithmetic, physiological, and agricultural.
- C. The method used to calculate population density reveals different information about the pressure the population exerts on the land.

Where is population distributed?

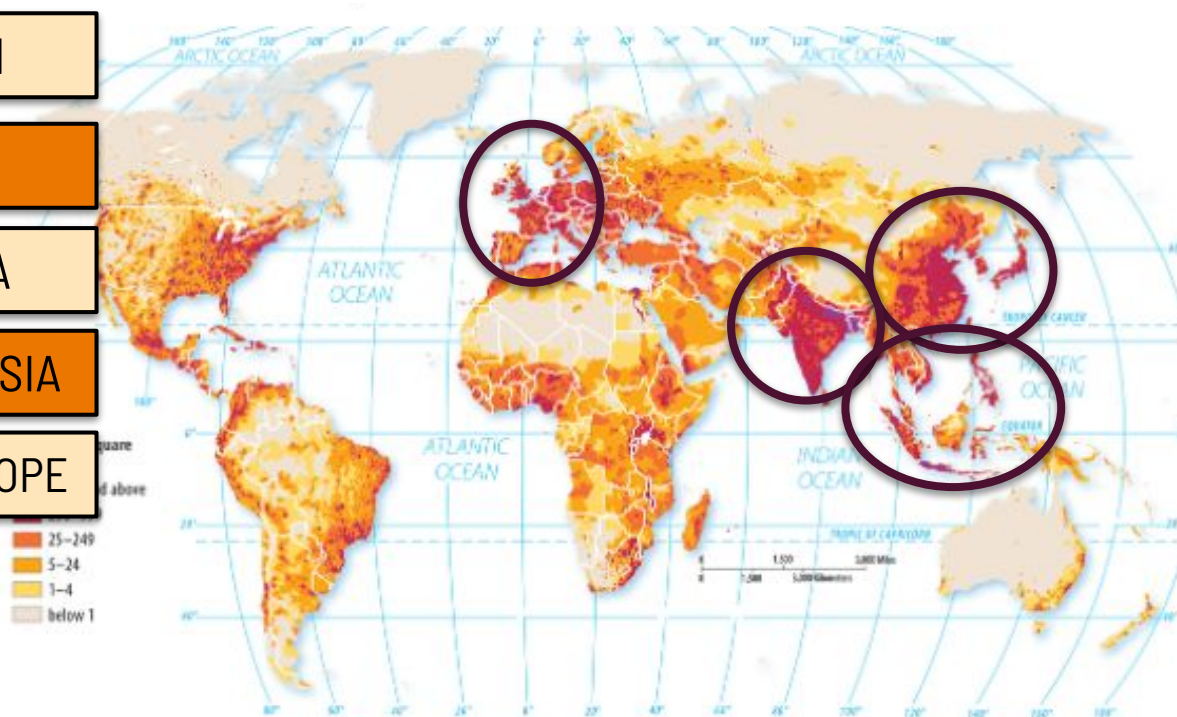
7.8 BILLION

EAST ASIA

SOUTH ASIA

SOUTHEAST ASIA

WESTERN EUROPE

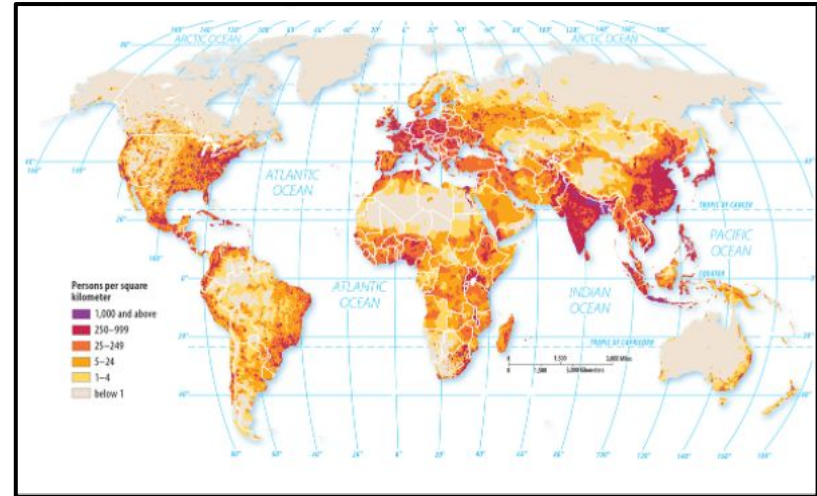


What factors influence the distribution of population?

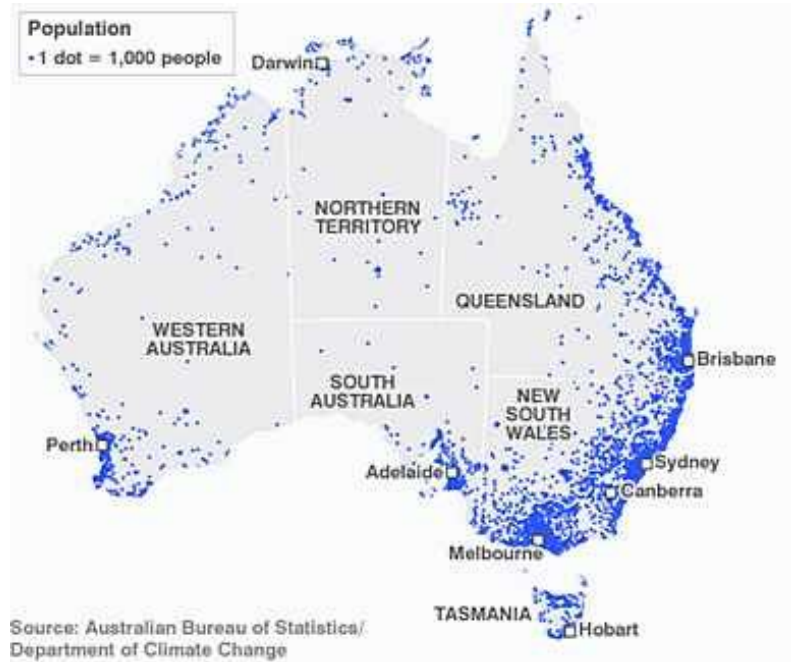


NATURAL / ENVIRONMENTAL / PHYSICAL

- People do not settle in the “5 toos”
 - Too hot, cold, wet, hilly, or dry
- People are likely to settle in...
 - Low elevation
 - Fertile soil
 - Temperate climates
 - Near a body of water
- Ecumene: The habitable areas of the world.



What factors influence the distribution of population?



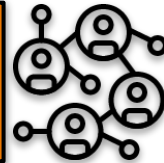
NATURAL / ENVIRONMENTAL / PHYSICAL

What OTHER factors influence the distribution of population?



ECONOMIC

SOCIAL / CULTURAL



POLITICAL

How do geographers calculate population density?

Arithmetic →

$$\frac{\text{Total population}}{\text{Land area (sq. mi)}}$$

Physiological →

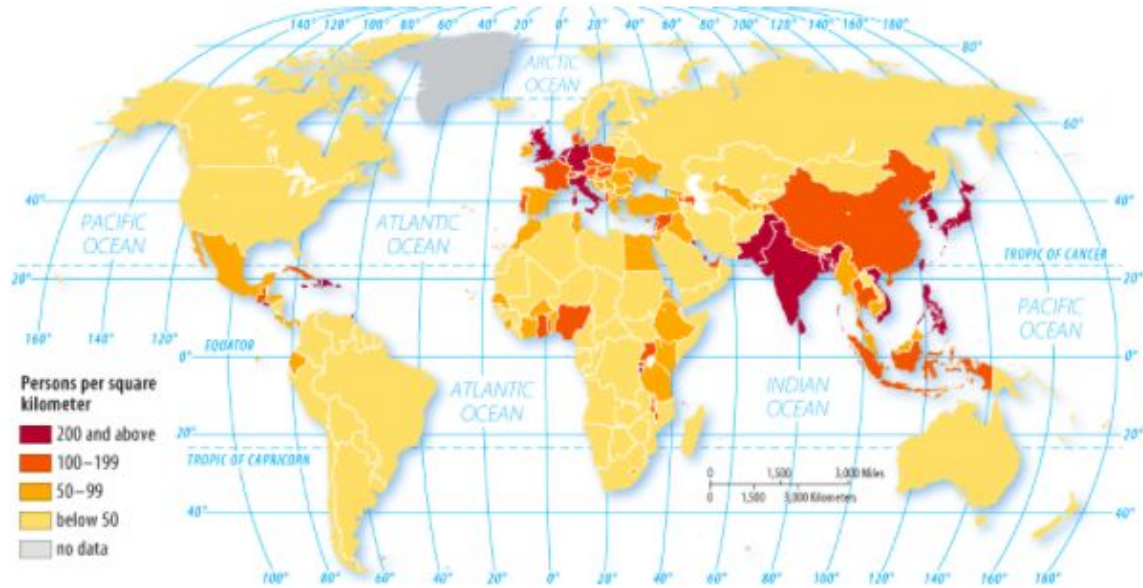
$$\frac{\text{Total population}}{\text{Arable Land (sq. mi)}}$$

Agricultural →

$$\frac{\text{Farmers}}{\text{Arable Land (sq. mi)}}$$

How do geographers calculate population density?

Does not account for the uneven distribution of population within each country. No information about clustering or dispersal.



Arithmetic

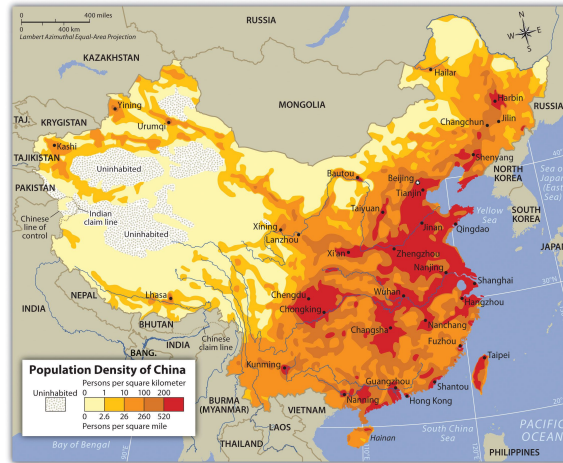


$$\frac{\text{Total population}}{\text{Land area (sq. mi)}}$$

How do geographers calculate population density?

How does our understanding of population distribution and density change when we change the scale of analysis?

94% of China's population lives in the Eastern region.

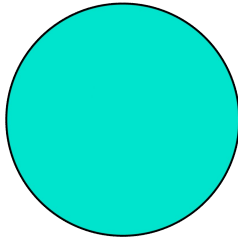


Arithmetic

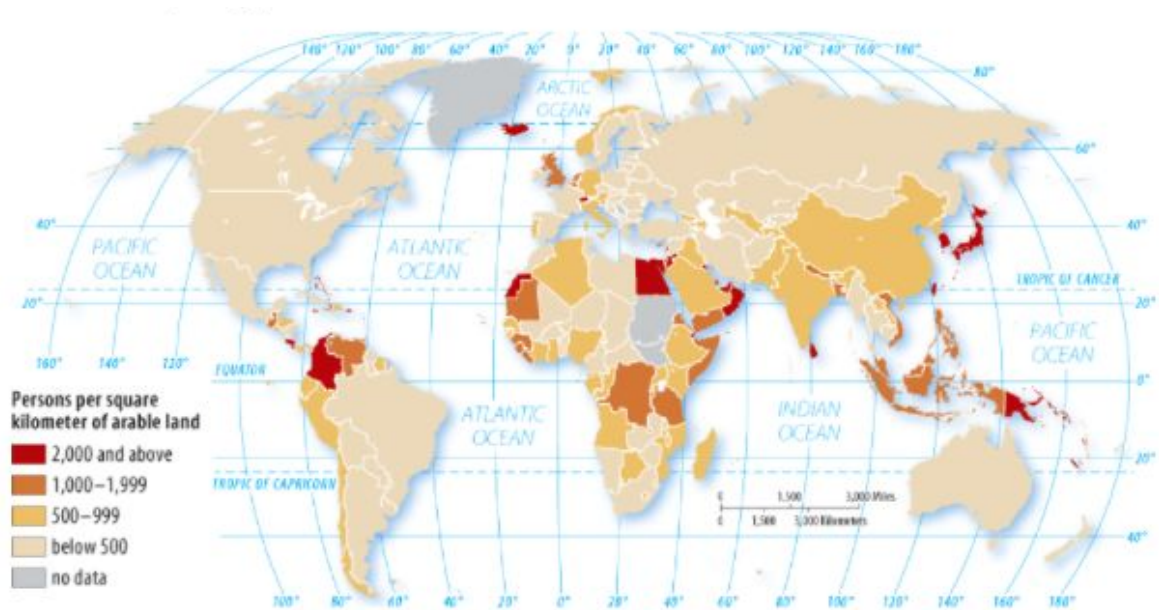


Total population
Land area (sq. mi)

How do geographers calculate population density?



Carrying Capacity: The maximum population size that can be supported by the available resources without damaging the environment or using natural resources unsustainably.



Physiological



$$\frac{\text{Total population}}{\text{Arable Land (sq. mi)}}$$

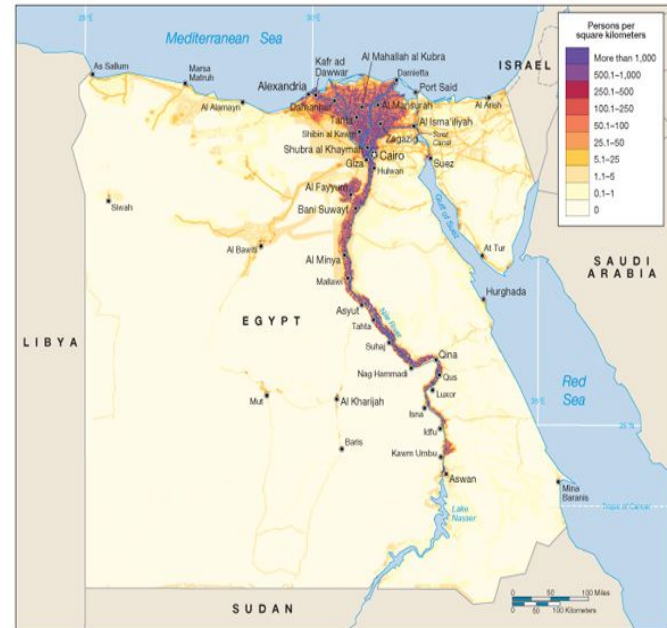
How do geographers calculate population density?

How does our understanding of population distribution and density change when use physiological density?

ARITHMETIC (2020) - 257

PHYSIOLOGICAL (2020) - 9,182

- 95% of the population of Egypt live on just 3% of land - along the Nile River.
- What is the physical environment of Egypt?



How do geographers calculate population density?

Better indicator of a country's level of **economic development** rather than population distribution.

High: LDCs

- Low level of technology
- Farming by hand = more farmers
- Low output

Low: MDCs

- High levels of mechanization
- Less farmers but more output.



Agricultural ➡

Farmers
Arable Land (sq. mi)

How do geographers calculate population density?

INTRODUCTION TO SUBSISTENCE FARMING



Agricultural ➡

Farmers
Arable Land (sq. mi)

Objective and Essential Learning

2.1.1 Identify the factors that influence the distribution of population at different scales.

2.1.2 Define and explain the differences between the methods geographers use to calculate population density.



What factors influence the distribution of population?



ECONOMIC

- Job Opportunities
- Access to Natural Resources



Case Study: Norilsk, Siberia

“One example of how economic conditions prevail over environmental limitations is Norilsk, Siberia. Nearly 200,000 people in this city, which is located north of the Arctic Circle, brave extremely harsh weather conditions ever winter. Why...The region has the world’s largest known deposits of nickel, an industrially important metal.”

Source: Human Geography - A Spatial Perspective, Bednarz, Bockenhauer & Hiebert



What factors influence the distribution of population?



SOCIAL / CULTURAL

- Housing availability
- Safety
- Access to transportation
- A feeling of belonging
- Community



POLITICAL

- Housing availability
- Safety
- Access to transportation
- A feeling of belonging
- Community